THE SAMPLE SIZE REQUIRED IN IMPORTANCE SAMPLING

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1 Introduction and contributions

1.1 Scientific Context

In Monte Carlo methods, and particularly in importance sampling, determining the required sample size is essential for reliable estimation. Importance sampling evaluates expectations under a target distribution ν using samples drawn from a proposal distribution μ, reweighted by the likelihood ratio $\rho = d\nu/d\mu$. When μ and ν are poorly aligned, especially in high-dimensional or rare-event regimes, the weights can be highly uneven, and a small number of samples may dominate the estimator.

Classical guidelines select the sample size based on variance considerations, since the variance typically decreases like $1/n$. However, empirical variance estimates may be misleading: a small observed variance does not guarantee accurate estimation when the weight distribution is highly skewed.

A central insight is that the required sample size has an information-theoretic nature: it scales essentially as $\exp\big(D(\nu\|\mu)\big)$, where $D(\nu\|\mu)$ denotes the Kullback–Leibler divergence. Thus, the intrinsic difficulty of importance sampling is governed by the entropic discrepancy between the target and proposal distributions, potentially leading to exponential sample complexity when they are far apart.

1.2 Problem Formulation

Let $(\mathcal{X}, \mathcal{F})$ be a measurable space and let μ and ν be probability measures with density ratio $\rho = \frac{d\nu}{d\mu}$. The objective is to compute $I(f) = \int_{\mathcal{X}} f(x)\, d\nu(x)$ for a measurable function f, assuming that direct sampling from ν is not available. Instead, we generate independent samples $X_1, \ldots, X_n \sim \mu$ and use the importance sampling estimator

$$I_n(f) = \frac{1}{n} \sum_{i=1}^{n} f(X_i)\rho(X_i),$$

which is unbiased.

The fundamental question is how large n must be for $I_n(f)$ to approximate $I(f)$ accurately. Since the weights $\rho(X_i)$ may be highly variable when μ and ν differ substantially, most samples can contribute negligibly while a few dominate the sum, causing instability. The core problem is therefore to characterize how the required sample size depends on the discrepancy between μ and ν, and to identify conditions under which importance sampling remains accurate and stable.

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2 Method of the article

2.1 Article Statement

Theorem 1 (Chatterjee–Diaconis, Theorem 1.1) *Let $(\mathcal{X}, \mathcal{F})$ be a measurable space, and let μ and ν be probability measures such that $\nu \ll \mu$, with density $\rho = \frac{d\nu}{d\mu}$. Let f be a measurable function and define*

$$I(f) = \int f \, d\nu, \qquad I_n(f) = \frac{1}{n} \sum_{i=1}^{n} f(X_i)\rho(X_i),$$

where $X_1, \ldots, X_n \sim \mu$ are i.i.d. Let $Y \sim \nu$ and set

$$L = D(\nu \| \mu) = \mathbb{E}_\nu[\log \rho(Y)].$$

If $n = \exp(L + t)$ for some $t \geq 0$, then

$$\mathbb{E}\big|I_n(f) - I(f)\big| \leq \|f\|_{L^2(\nu)} \left(e^{-t/4} + 2\sqrt{\mathbb{P}\big(\log \rho(Y) > L + t/2\big)} \right).$$

Conversely, if $n = \exp(L - t)$ for some $t \geq 0$, then for any $\delta \in (0, 1)$,

$$\mathbb{P}\big(I_n(1) \geq 1 - \delta\big) \leq e^{-t/2} + \frac{\mathbb{P}\big(\log \rho(Y) \leq L - t/2\big)}{1 - \delta}.$$

Theorem 1.1 is much stronger than a variance-based approach because it gives *matching* (up to fluctuations of $\log \rho(Y)$) sufficient *and* necessary conditions on the sample size n for importance sampling to be accurate, and it does so in a way that is intrinsically tied to the discrepancy between μ and ν through $L = D(\nu \| \mu)$. In contrast, a classical practical diagnostic is to estimate the variance via the plug-in (empirical) estimator

$$v_n(f) \;:=\; \frac{1}{n^2} \sum_{i=1}^{n} f(X_i)^2 \rho(X_i)^2 \;-\; \frac{I_n(f)^2}{n},$$

and to declare convergence whenever $v_n(f)$ falls below a tolerance ε. The issue is that when μ and ν are poorly matched, $\rho(X_i)$ is typically extremely small under μ (the rare large weights that matter may not have appeared yet), so the empirical second moment $\frac{1}{n^2} \sum f(X_i)^2 \rho(X_i)^2$ can look artificially tiny, making $v_n(f)$ suggest "low variance" even though the estimator is still far from the target. This is precisely formalized by Theorem 2.1: it shows that the event $\{v_n(f) < \varepsilon\}$ can occur with high probability at a sample size that depends essentially only on ε and *not* on the difficulty of the pair (μ, ν), meaning the diagnostic can be almost universally over-optimistic.

Theorem 2 (Chatterjee–Diaconis, Theorem 2.1) *Given any $\varepsilon > 0$, there exists an integer $n \leq \varepsilon^{-2}\, 2^{1+\varepsilon^{-3}}$ such that the following holds. For any probability measures μ, ν with $\nu \ll \mu$, any measurable f with $\|f\|_{L^2(\nu)} \leq 1$, and i.i.d. samples $X_1, \ldots, X_n \sim \mu$, if $v_n(f)$ is defined by*

$$v_n(f) \;:=\; \frac{1}{n^2} \sum_{i=1}^{n} f(X_i)^2 \rho(X_i)^2 \;-\; \frac{I_n(f)^2}{n},$$

then

$$\mathbb{P}\big(v_n(f) < \varepsilon\big) \;\geq\; 1 - 4\varepsilon.$$

Taken together, Theorem 1.1 and Theorem 2.1 explain the gap: Theorem 1.1 identifies the *correct scale* for convergence (typically $n \approx e^{D(\nu\|\mu)}$ under concentration), whereas Theorem 2.1 shows that a common variance-estimation stopping rule can declare convergence at an n that is essentially unrelated to $D(\nu\|\mu)$, so it cannot reliably detect the hard regimes where importance sampling truly requires exponentially many samples.

2.2 Limitations

Although Theorem 1.1 provides a sharp exponential threshold $n \approx \exp(L)$ with $L = D(\nu\|\mu)$, computing L in practice is typically difficult. In Example 3.3 (independent spins), we have $G_\beta(x) = \exp(-\beta H(x))/Z(\beta)$ and $L = F(\beta_0) - F(\beta) - (\beta_0 - \beta)F'(\beta)$ with $F(\beta) = \log Z(\beta)$. Thus, even in this simple exponential family setting, L depends on the log-ratio of normalizing constants $\log Z(\beta_0) - \log Z(\beta)$.

While $Z(\beta)$ is explicit for independent spins, in most realistic high-dimensional models the partition function $Z(\beta)$ is intractable. Estimating L then amounts to estimating the ratio $Z(\beta)/Z(\beta_0)$, which is itself a computationally demanding task. A standard approach is thermodynamic integration, using $\log \frac{Z(\beta)}{Z(\beta_0)} = -\int_{\beta_0}^{\beta} \mathbb{E}_t[H]\, dt$, but this requires simulating from G_t for many intermediate values of t, typically via MCMC. The resulting estimate is subject to discretization error and Monte Carlo error, and may be unreliable if mixing is slow.

Moreover, the upper bounds in Theorem 1.1 are not optimized for tight finite-sample guarantees; their role is to match the order of magnitude of the lower bound and establish the exponential scale $\exp(D(\nu\|\mu))$, rather than to provide sharp constants or practically usable error bars. In particular, the results rely on concentration properties of $\log \rho(Y)$ under ν, which may fail or be difficult to verify in complex models.

2.3 Solutions proposed

Although Theorem 1.1 shows that the critical sample size is of order $\exp(D(\nu\|\mu))$, this quantity depends explicitly on the target measure ν, which is typically intractable in practice. Since $D(\nu\|\mu) = \mathbb{E}_\nu[\log \rho(Y)]$ cannot be computed without detailed knowledge of ν, practitioners instead rely on variance-based diagnostics. Using the fact that $\mathrm{Var}(I_n(f))$ scales like $1/n$, one estimates the variance empirically via $v_n(f) = \frac{1}{n^2}\sum_{i=1}^n f(X_i)^2\rho(X_i)^2 - \frac{I_n(f)^2}{n}$ and declares convergence when $v_n(f)$ becomes small. However, Theorem 2.1 shows that this strategy can be misleading: with high probability, $v_n(f)$ may fall below any fixed tolerance at a sample size that does not reflect the intrinsic difficulty of the pair (μ, ν). The reason is that under μ, most weights $\rho(X_i)$ are typically extremely small, so the empirical second moment appears tiny long before the rare but decisive large weights have been observed.

To address this issue, the authors propose a diagnostic based on weight concentration rather than second moments. Let $X_1, \ldots, X_n \sim \mu$ and define $Q_n = \frac{\max_{1\le i\le n}\rho(X_i)}{\sum_{i=1}^n \rho(X_i)}$ and $q_n = \mathbb{E}(Q_n)$. The statistic Q_n measures how much a single weight dominates the total mass: if $n \ll e^{D(\nu\|\mu)}$ (under suitable concentration), typically one observation carries a non-negligible fraction of the total weight, so Q_n remains bounded away from zero, whereas when the estimator has stabilized, Q_n becomes small. Moreover, Theorem 2.3 shows that smallness of q_n is a necessary condition for convergence: if $\varepsilon_n = \mathbb{E}|I_n(1) - 1|$, then $q_n \le C\max\{\frac{1}{n}, \frac{\log\log(1/\varepsilon_n)}{\log(1/\varepsilon_n)}\}$, so true accuracy forces q_n to be small. Thus, unlike empirical variance, Q_n is theoretically linked to the genuine convergence scale while remaining computable from the simulated weights alone.

3 Experimental setup

We conduct several numerical experiments to illustrate the theoretical prediction that importance sampling requires a sample size of order $\exp(L)$ where $L = D(\nu\|\mu)$, and to compare diagnostics based on the empirical variance $v_n(1)$ and the weight dominance statistic $Q_n = \max_i w_i / \sum_i w_i$.

First, we consider a Gaussian shift model with $\mu = \mathcal{N}(m, I_d)$ and $\nu = \mathcal{N}(0, I_d)$ where $m = (a, \ldots, a) \in \mathbb{R}^d$. In this case the likelihood ratio is explicit and $L = \frac{1}{2}da^2$. For several values of a and sample sizes n, we estimate $I_n(1)$ together with the diagnostics $v_n(1)$ and Q_n, and record the median relative error across repeated runs.

Second, we investigate a setting in which $\log \rho$ is less concentrated by considering a Gaussian mixture target of the form $\nu = (1-\varepsilon)\mathcal{N}(-m, I_d) + \varepsilon\mathcal{N}(+m, I_d)$ with proposal $\mu = \mathcal{N}(0, I_d)$. This construction introduces multimodality and, when ε is small, a rare but influential component. Since $L = D(\nu\|\mu)$ does not admit a closed-form expression in this setting, we estimate it via Monte Carlo

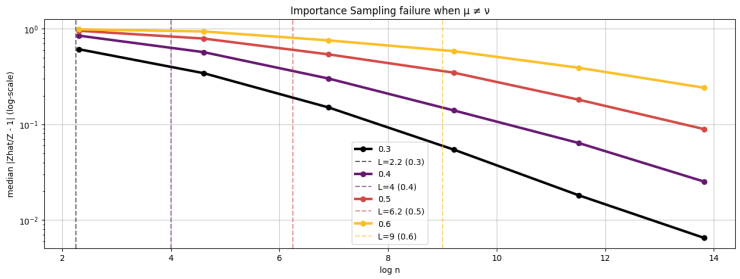
using $L \approx \mathbb{E}_\nu[\log(d\nu/d\mu)]$. We then repeat the same importance sampling experiments, comparing empirical performance with the theoretical threshold based on L, in order to assess whether L remains informative when $\log \rho$ exhibits weaker concentration and heavier tails.

Finally, we consider the Gibbs framework using the independent-spin model with Hamiltonian $H(x) = -\sum_i x_i$. In this case the theoretical quantities $L = F(\beta_0) - F(\beta) - (\beta_0 - \beta)F'(\beta)$ and $\sigma = |\beta - \beta_0|\sqrt{F''(\beta)}$ are available. For several system sizes N, we generate samples from G_{β_0} either i.i.d. or via Gibbs sampling, choose sample sizes according to $\log n = L + c\sigma$, and evaluate the estimator of $Z(\beta)$ together with the diagnostics $v_n(1)$ and Q_n.

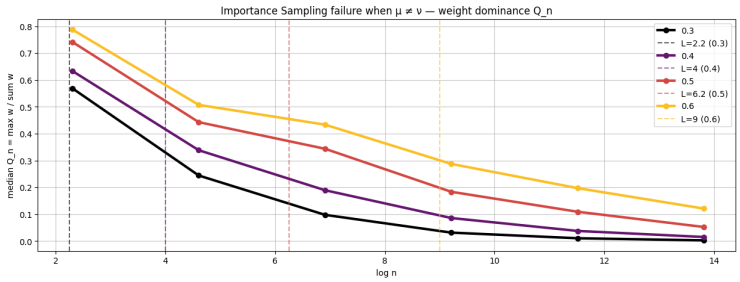
In addition, we study the practical setting where L is unknown by estimating it via thermodynamic integration using $F(\beta) - F(\beta_0) = -\int_{\beta_0}^{\beta} \mathbb{E}_t[H]dt$, which yields an estimator $\widehat{L}$, and repeating the experiments with $n \approx \exp(\widehat{L} + c\sigma)$.

4 Results

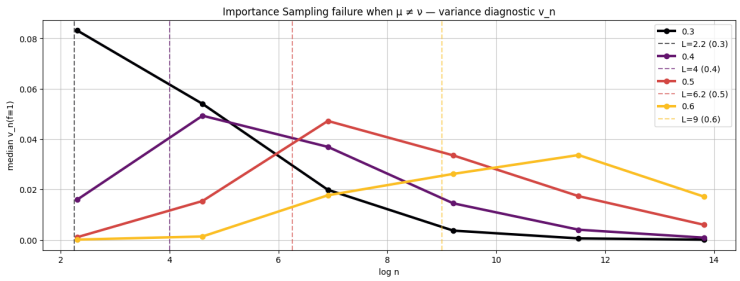
4.1 Ideal Case



(a) Relative error of the IS estimator



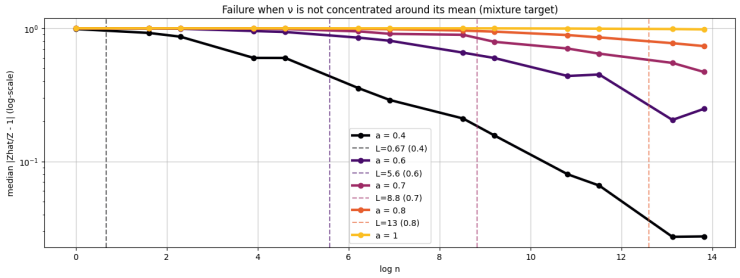
(b) Weight degeneracy diagnostic Q_n



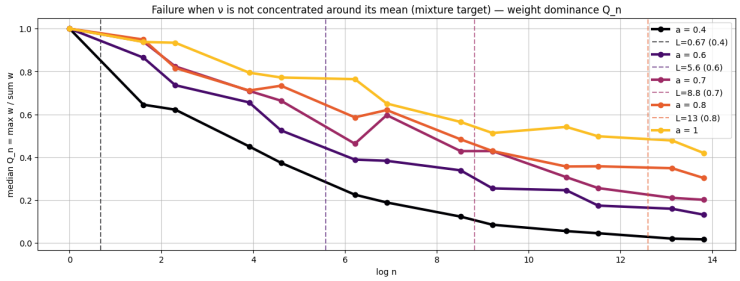
(c) Empirical variance proxy $v_n(1)$

Figure 1: Gaussian shift experiment with $d = 50$. We consider $a \in \{0.3, 0.4, 0.5, 0.6\}$, sample sizes $n \in \{10^1, \ldots, 10^6\}$, and $R = 200$ replications per setting. The figure shows the relative error, weight degeneracy diagnostic Q_n, and variance proxy. The results confirm the relevance of the threshold $L = \frac{1}{2}da^2$ and demonstrate that Q_n is a more reliable indicator of failure than $v_n(1)$.

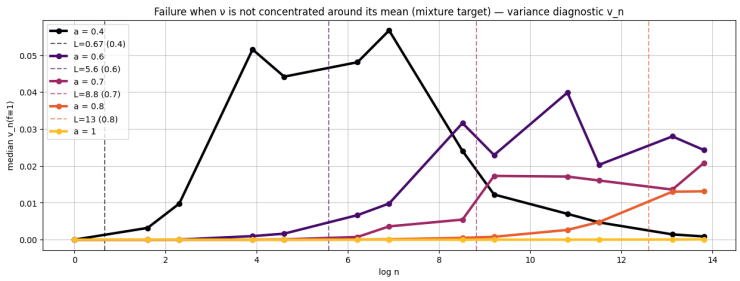
4.2 Rare Event



(a) Relative error of the IS estimator



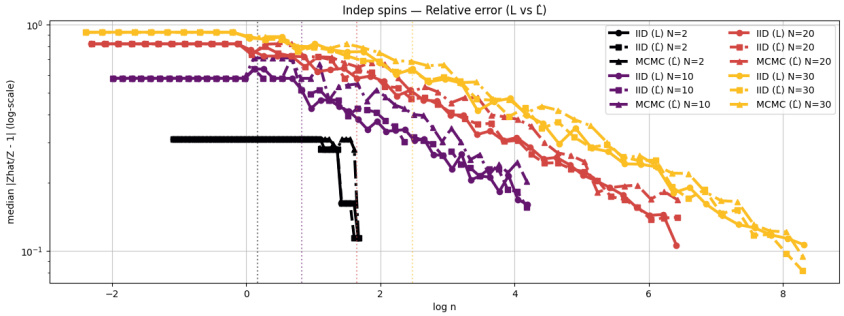
(b) Weight degeneracy diagnostic Q_n



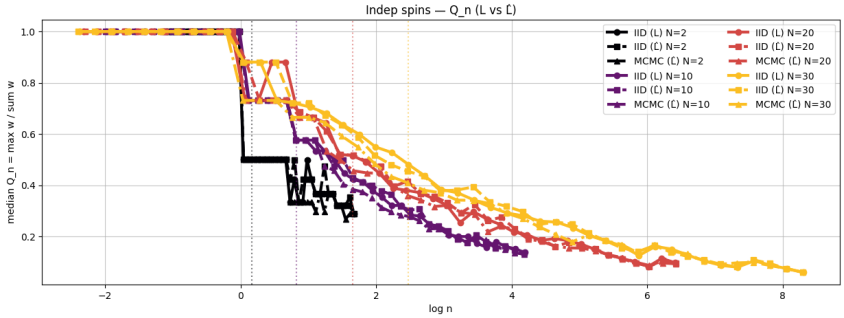
(c) Empirical variance proxy $v_n(1)$

Figure 2: Rare-event mixture experiment with dimension $d = 50$. The target distribution is bimodal, with shift parameters $a \in \{0.4, 0.6, 0.7, 0.8, 1\}$. The sample sizes are $n \in \{10^0, 5\times10^0, 10^1, 5\times10^1, \ldots, 10^6\}$, each configuration is replicated $R = 100$ times, and L is estimated using $M_L = 30\,000$ samples and $\epsilon = 0.001$. The figure reports the relative error, the weight degeneracy diagnostic Q_n, and the variance proxy $v_n(1)$. In contrast with the Gaussian shift case, the theoretical threshold based on L is less informative: despite large sample sizes relative to e^L, the estimator may still fail due to the lack of concentration of the target distribution. This illustrates that when ν is not concentrated around its mean, the KL-based threshold no longer sharply predicts the success of importance sampling, while Q_n remains a more reliable indicator of weight degeneracy.

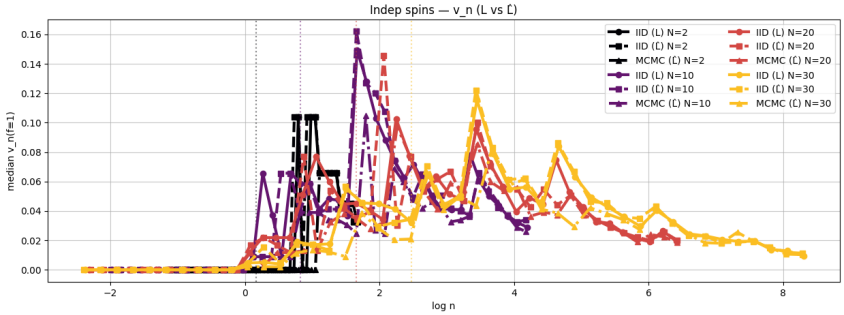
4.3 Independent Spins



(a) Relative error of the IS estimator



(b) Weight degeneracy diagnostic Q_n



(c) Empirical variance proxy $v_n(1)$

Figure 3: Independent spins experiment with $N \in \{2, 10, 20, 30\}$, $\beta_0 = 1$, $\beta = 0.5$, and $c \in [-2, 2.5]$ (step 0.1), with 200 replications per configuration. The sample size is chosen according to $\log n = L + c\sigma$, and we compare IID and MCMC sampling, as well as the use of the theoretical L versus its estimator $\widehat{L}$. The three panels report the relative error, the degeneracy diagnostic Q_n, and the variance proxy $v_n(1)$. Across all system sizes, the qualitative behavior is essentially identical: the phase transition around $\log n \approx L$ is preserved, and using MCMC instead of IID sampling or replacing L by $\widehat{L}$ does not significantly alter the threshold. This confirms the robustness of the e^L scaling prediction in the independent spins setting. (It could be because of the simplicity of the ind. spins model)